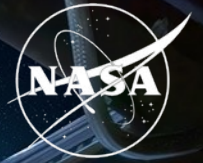


National Aeronautics and
Space Administration



Synthetic transcriptomics for mouse spaceflight

Can generated RNA-seq improve FLT vs GC
analysis within each tissue?

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QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

1,610

biological profiles

75

accessions

835 FLT

775 GC

ARCHS4 mouse

997,515

profiles audited

17,244

selected

20 tissues

- Mission, strain, material and assay differences can resemble a flight effect.
- The generator must preserve tissue and FLT/GC structure without memorizing samples.
- Statistical evidence must still come from observed OSDR profiles.

METHOD

We built a configurable bulk RNA-seq generation pipeline

Data scope, transforms, harmonization, training and conditioning can change without changing the evaluation contract.

Data + scope

OSDR API or ARCHS4
one or many studies
pooled or per tissue

Transform

raw / CPM / TPM
log + scaling
all genes / HVG / L1000

Harmonize

none or study z
ComBat / MBatch
MOBER adapter

Model + train

WGAN-GP or DDIM
OSDR or ARCHS4
pretrain + adapt

Condition

FLT/GC + tissue
study + material
available covariates

Selected branch **TPM / MaxAbs -> ARCHS4 DDIM -> OSDR adaptation.** Conditions: tissue, FLT/GC, accession and material; no global batch correction.

A 463-row experiment plan and nine matched liver harmonization arms were screened before full training.

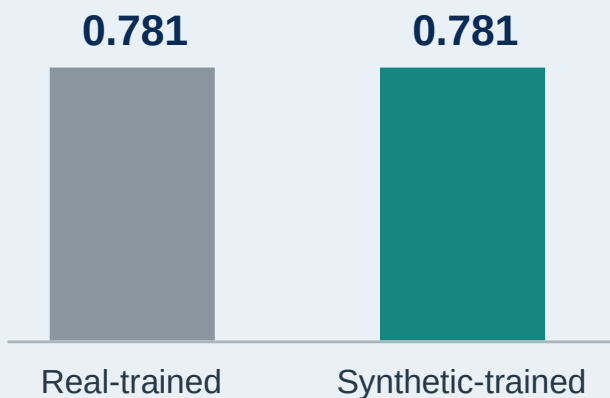
VALIDATION

DDIM matched expression and reduced separability

AA closer to 0.5 means a classifier has less success separating real and generated profiles; lower FD is better.

ARCHS4 tissue probe

Balanced accuracy on held-out real profiles



Metric	WGAN-GP	DDIM	Reading
Correlation	0.976	0.974	similar
F1	0.985	0.997	higher
Adversarial accuracy	0.636	0.475	closer to 0.5
FD / real P95	0.144	0.074	lower

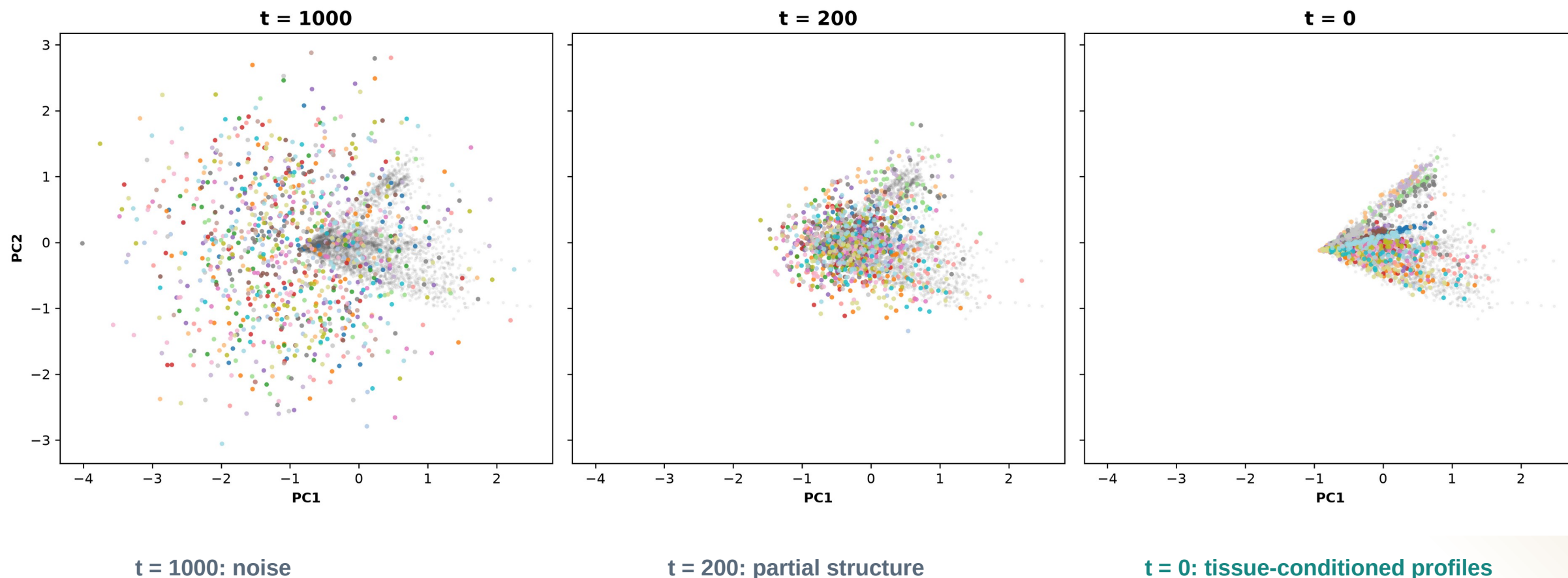
**Use DDIM
downstream**

Lower separability and distributional distance, with comparable correlation and higher F1.

DIFFUSION

Diffusion learns tissue structure from noise

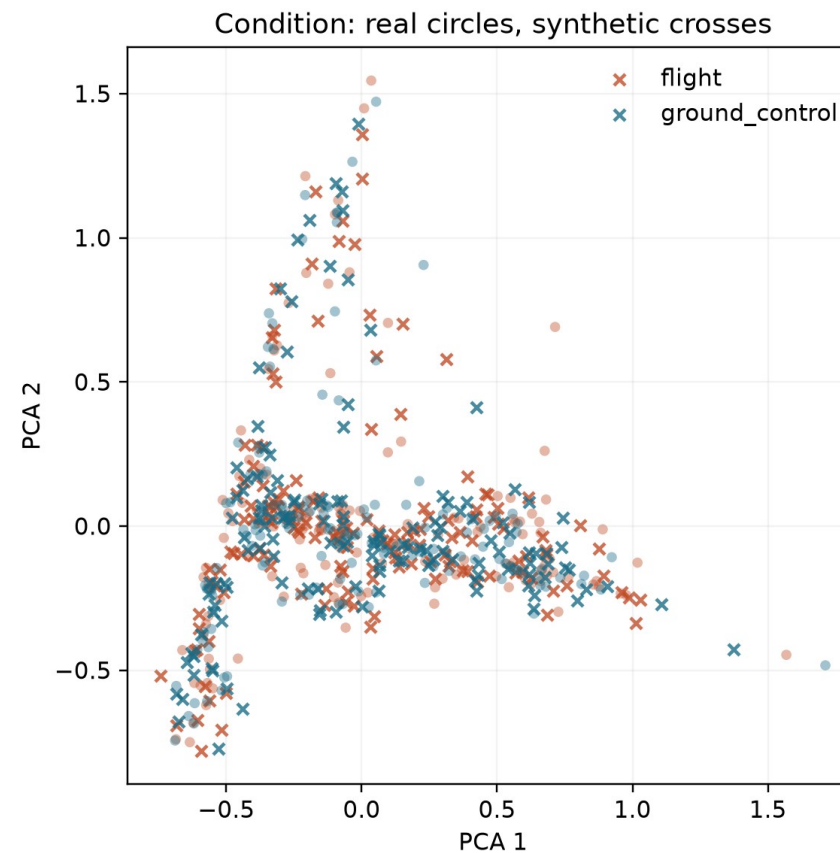
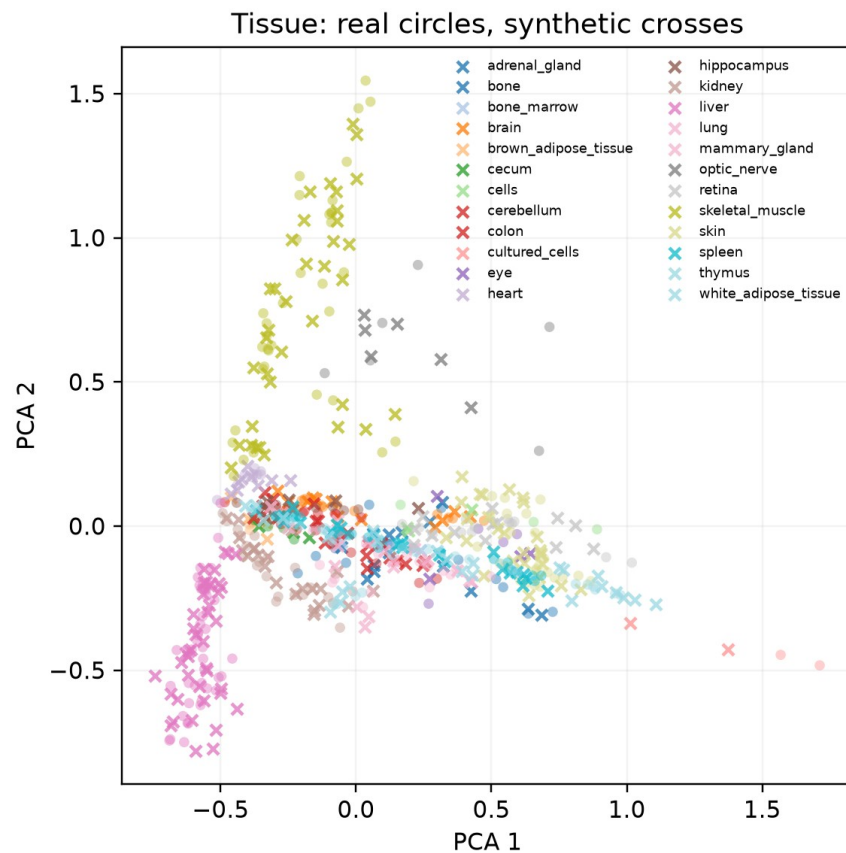
The same PCA axes follow 1,024 generated profiles through reverse diffusion.



DIFFUSION OUTPUT

Generated profiles track the real OSDR PCA manifold

Circles are locked real OSDR profiles; crosses are matched DDIM profiles from generation seed 5020.

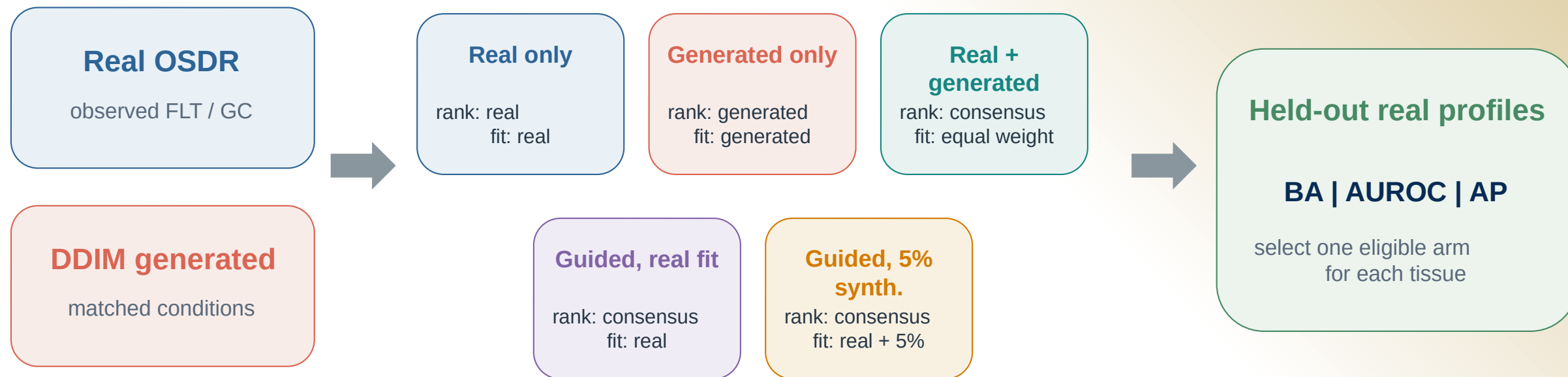


Interpretation: tissue structure dominates; FLT and GC are subtler. PCA is descriptive, so quantitative metrics determine validation.

ANALYSIS

Synthetic profiles entered the analysis in five different ways

Every arm was judged on held-out real profiles before synthetic attribution was allowed.



Biology check FLT vs GC effects, random-effects meta-analysis and BH FDR use real OSD R profiles only.

Synthetic profiles never increase animal n.

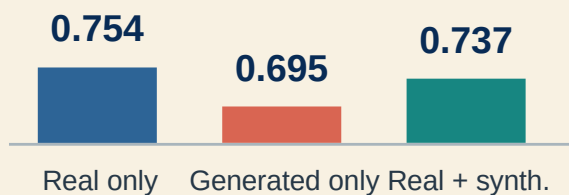
UTILITY

Pooling tissues hid useful signal

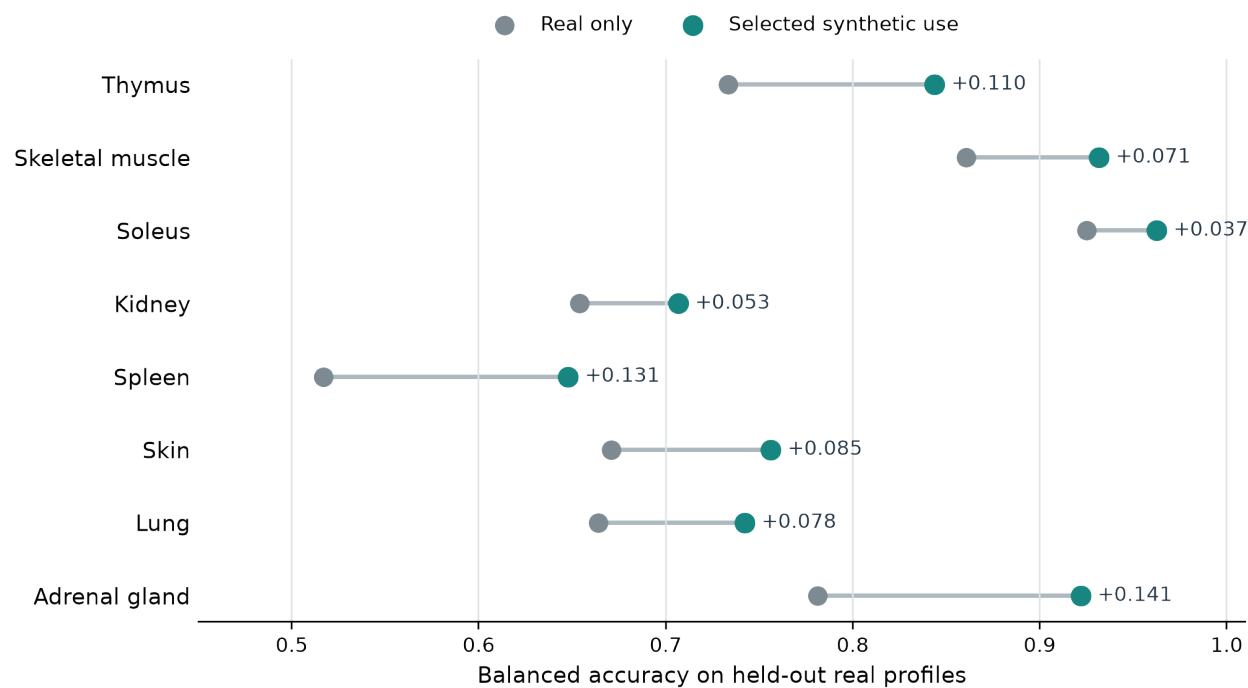
A single FLT/GC classifier did not improve, but tissue-specific synthetic use often did.

Pooled across tissues

Balanced accuracy



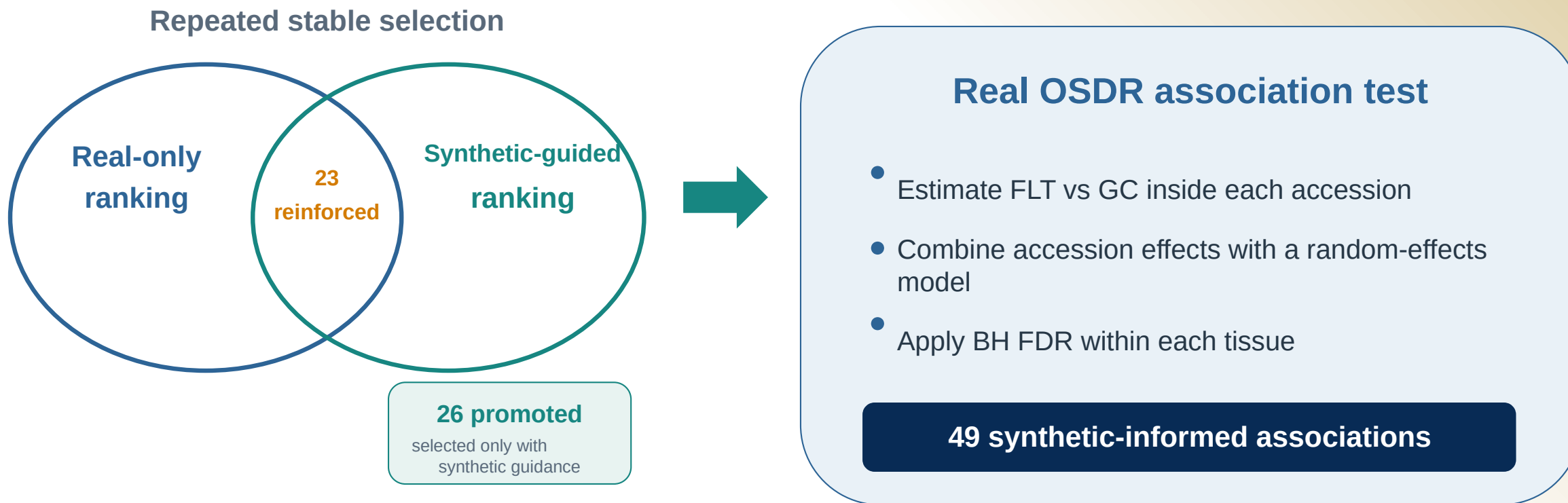
Selected use within each tissue



The useful arm differed by tissue.

Synthetic guidance changed ranking, not statistical evidence

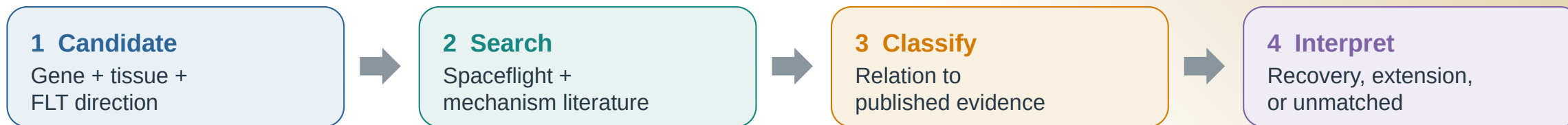
Promoted and reinforced describe repeated feature selection; both require a supporting effect in real OSD data.



These are 49 tissue-gene associations among 459 BH-FDR rows. A gene can appear in more than one tissue.

Selection and literature are separate dimensions

All 49 synthetic-informed associations received both a selection status and a literature interpretation.



Selection status

26

Promoted

Stable only with synthetic guidance

23

Reinforced

Stable with and without guidance

Literature interpretation

22

Aligning

Direct or same-tissue process agreement

19

Complementary

Related process or mechanism

4

Ambiguous

Mixed or context-dependent evidence

4

Unmatched

No sufficiently specific match found

A reinforced gene can be complementary, ambiguous, or unmatched; an aligning gene can be promoted.

ANALYSIS COVERAGE

All 27 completed tissue analyses were retained

The biological narrative narrows only after reporting synthetic-informed, real-only, and null outcomes.

10

Synthetic-informed

Promoted or reinforced BH-FDR gene

Adrenal Gland
Eye
Kidney
Skeletal muscle (pooled)
Skin
Spleen
Thymus
Gastrocnemius
Soleus
Tibialis Anterior

5

Real BH FDR only

Real association, no synthetic-informed selection

Heart
Liver
Retina
EDL
Quadriceps

12

No panel BH-FDR gene

No BH-FDR gene in the 974-gene panel

Bone
Bone Marrow
Brain
Brown Adipose Tissue
Cecum
Cerebellum
Colon
Hippocampus
Lung
Mammary Gland
Optic Nerve
White Adipose Tissue

GENE INVENTORY

Synthetic-informed genes spanned 10 tissue analyses

All 49 associations passed BH FDR in real OSD data; synthetic profiles affected prioritization, not the statistical test.

P: promoted (bold) ■ aligning ■ complementary ■ ambiguous ■ unmatched

R: reinforced (regular)

Color = literature | P/R = selection

Thymus
16 genes

FLT higher *P:Hsd17b11, P:Etv1, R:Snx7*

FLT lower *P:Nusap1, P:Stmn1, **P:Birc5**, P:Cdk1, P:Top2a, P:Ccnb2, P:Aurka, P:Ccne2, P:Kif20a, P:Pcna, P:Ccnf, R:Ube2c, R:Gmnn*

Spleen
4 genes

FLT higher *P:Rai14, P:Myl9, P:Ptprk, R:Loxl1*

FLT lower *none*

Kidney
2 genes

FLT higher *P:Inpp4b, R:Slc37a4*

FLT lower *none*

Gastrocnemius
2 genes

FLT higher ***P:Nfkbia***

FLT lower ***P:Fhl2***

Eye
1 gene

FLT higher *none*

FLT lower *R:Klhl21*

Skeletal muscle (pooled)
12 genes

FLT higher *R:Sox4, R:Cebpd, R:Sh3bp5, R:Prkcd, R:Arid5b, R:Sesn1, R:Tle1*

FLT lower *P:Klhl21, P:Mapkapk5, P:Reep5, **P:Itgb5**, R:Bphl*

Soleus
5 genes

FLT higher *R:Tpm1*

FLT lower *R:Bdh1, R:Ech1, **R:Bnip3**, R:Decor1*

Tibialis anterior
4 genes

FLT higher *P:Cebpd, **R:Cdkn1a**, R:St3gal5, R:Bnip3*

FLT lower *none*

Adrenal gland
2 genes

FLT higher *none*

FLT lower ***P:Psmb8**, R:Tspan4*

Skin
1 gene

FLT higher ***P:Plscr1***

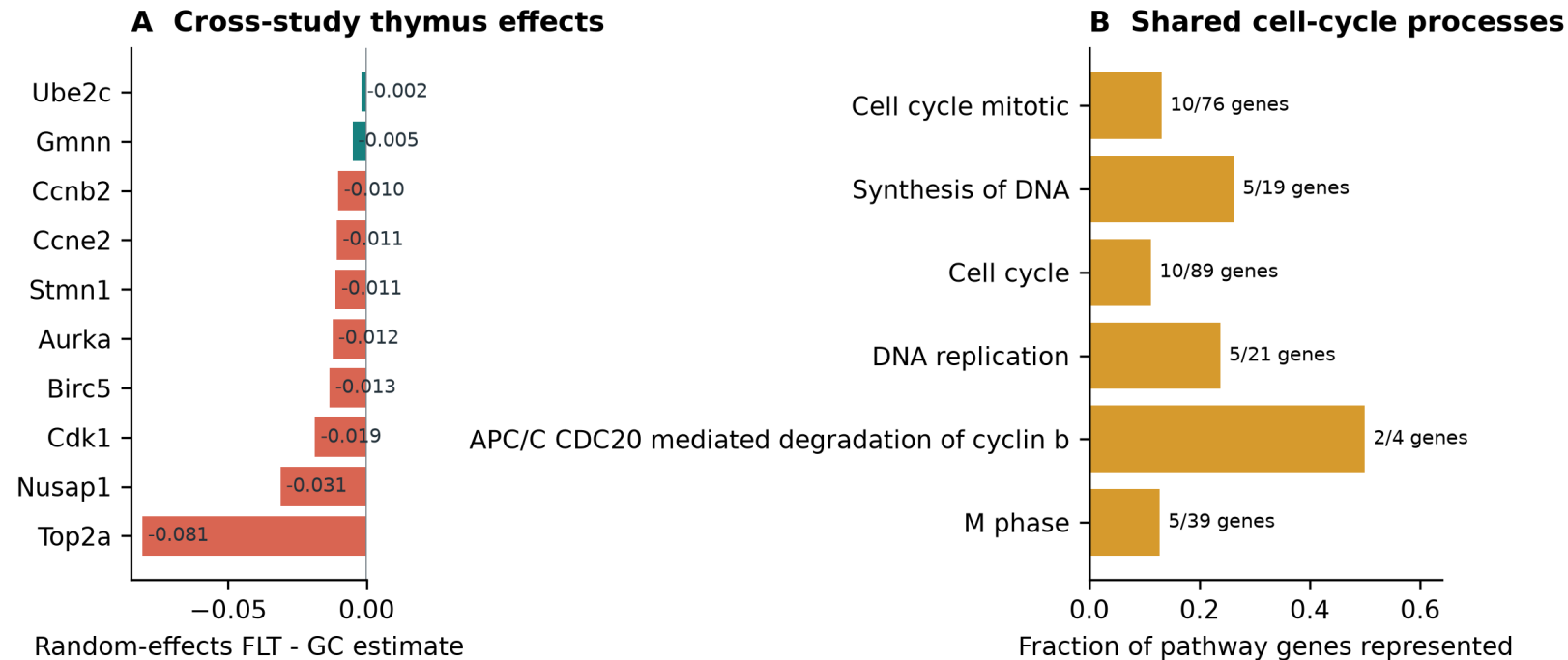
FLT lower *none*

THYMUS

Thymus points to lower proliferative renewal

The core panel recovers cell-cycle biology; two flight-higher genes extend the hypothesis.

Synthetic-informed thymus genes converge on a flight-lower mitotic program



16 synthetic-informed
BH-FDR associations

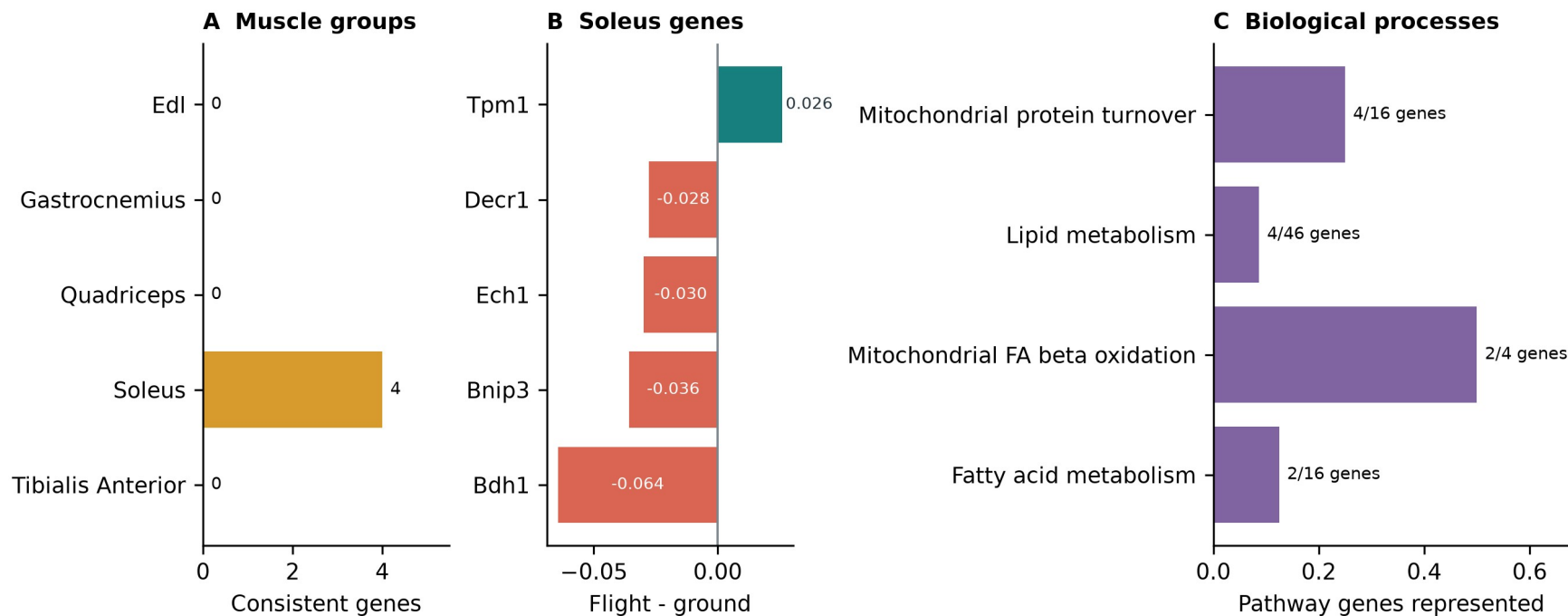
13 promoted | 3 reinforced

- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Higher Hsd17b11 and Etv1 suggest lipid or T-cell-state shifts
- Bulk RNA-seq cannot separate cell loss from lower transcription

Soleus reinforces a mitochondrial and lipid program

This result strengthens an existing real-data pattern rather than promoting a new soleus gene.

Anatomical separation reveals a soleus-specific oxidative-metabolism hypothesis



0.925 -> 0.963

balanced accuracy

5 reinforced | 0 promoted

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling

ADDITIONAL FINDINGS

Additional tissues produced narrower gene-level hypotheses

Thymus and soleus remained the strongest coherent programs, but they were not the only synthetic-informed results.

Pooled muscle	12 genes: stress, differentiation, mitochondrial and signaling candidates	Mixed panel
Kidney + spleen	2 renal genes and 4 splenic adhesion/ECM candidates	Gene-level
Skin + eye	Plscr1 higher in skin; Klhl21 lower in eye	Gene-level
Other muscle groups	Gastrocnemius 2 genes; tibialis anterior 4 genes	Gene-level
Adrenal gland	Psmb8 and Tspan4 were lower in flight and literature unmatched	Unmatched

These findings remain available for follow-up without displacing the stronger thymus and soleus programs.

TAKEAWAYS

Synthetic data helped most as a tissue-specific prior

The generator created useful structure, not new biological replication.

1 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

2 Use

Pooled augmentation failed. Tissue-specific ranking and low-weight training were more useful.

3 Interpret

Thymus and soleus were clearest. Literature review separated recovery from complementary hypotheses.

Next test

Use independent samples and cell-resolved assays to test thymus proliferation, soleus metabolism and prioritized candidates from additional tissues.

Generated profiles never enter the biological sample count or the BH-FDR test.

Thank you

Questions?

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Mentor

James Casaletto

Program

NASA Space Life Sciences Training Program

Data

NASA OSDR | ARCHS4 | Reactome

Compute

NASA Ames Research Center

Code and
manuscript

github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSDR profiles.